電子學實驗 (一)
Integrators, Differentiators, and Their Frequency Responses
結報

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1 Exp. 1 Integrator

I Circuit

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input resistance $R = 1$ kΩ(current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a capacitance $C = 0.01$ μF (103) (current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

i Theoretical results

$v_O(0^-)$ is the initial voltage across C (v_O relative to v_-).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. 90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Zero-state responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

$$v_O(t) = -\frac{AT}{RC}\left(2s\,|s| - s\right), \quad s = \frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Triangle wave output with amplitude $\frac{AT}{4RC}$

$$v_O(t) = -\frac{AT}{RC}\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right)$$

- Sinusoidal input:

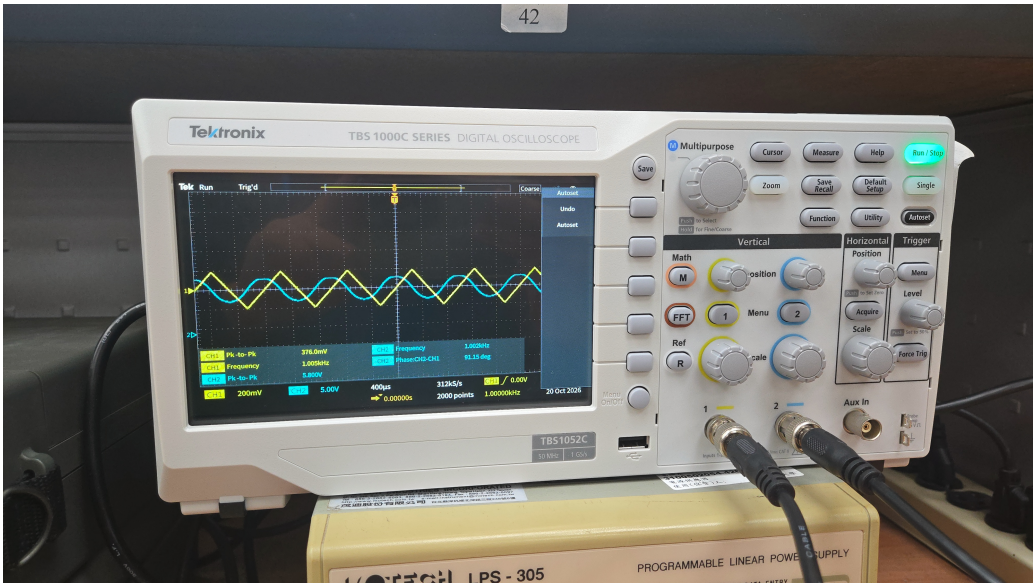
$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

Sinusoidal output 90 degrees leading input with amplitude $\frac{AT}{RC}$.

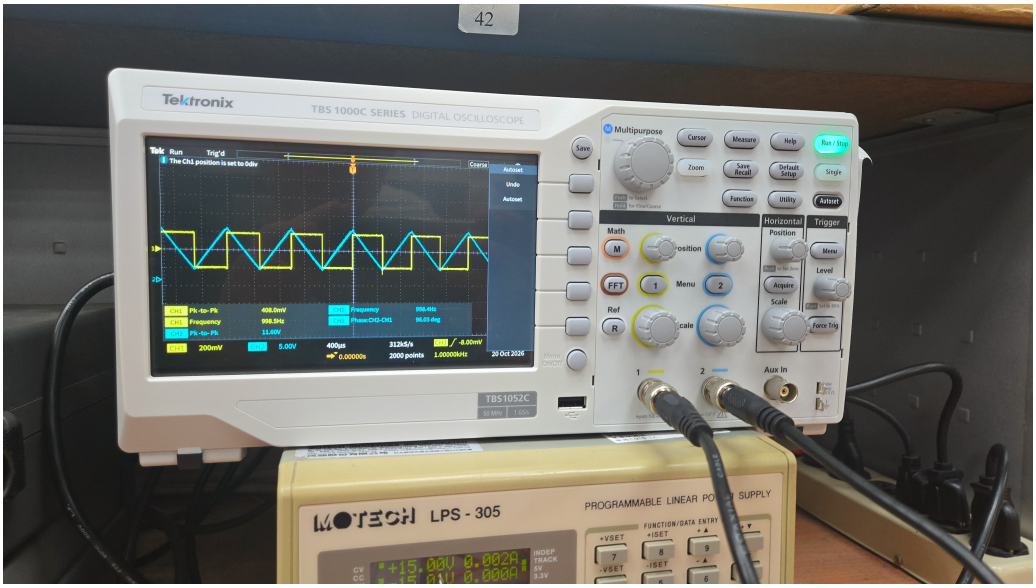
$$v_O(t) = \frac{AT}{RC}\cos\left(\frac{t}{T}\right)$$

II 實驗結果

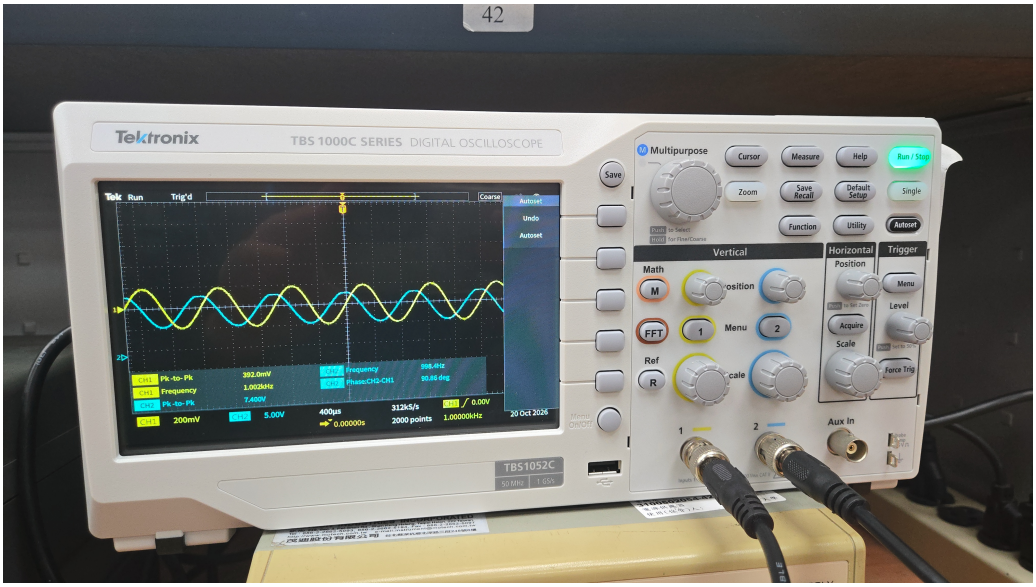
Triangle wave:



Square wave:



Sinusoidal wave:



2 Exp. 2 Differentiator

I Circuit

Measure v_I and v_O waveforms of an op amp where input voltage v_I is connected to v_- through an input capacitor $C = 0.01$ µF (103) (current from v_I to v_-), v_+ is connected to ground, and v_O is fed back to v_- through a resistor $R = 1$ kΩ(current from v_- to v_O), under triangle, square, and sinusoidal v_I with amplitude 0.2.

II Theoretical results

$v_I(0^-)$ is the initial voltage across C (v_I relative to v_-).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. -90 degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

  Square wave with amplitude $\frac{4ARC}{T}$

$$v_O(t) = -\frac{8ARC}{T}\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad \frac{2t}{T} \notin \mathbb{Z}$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Dirac delta sum output

$$v_O(t) = 2ARC\sum_{n\in\mathbb{Z}} \delta(t - nT) - \delta(t - \left(n + \frac{1}{2}\right)T)$$

- Sinusoidal input:

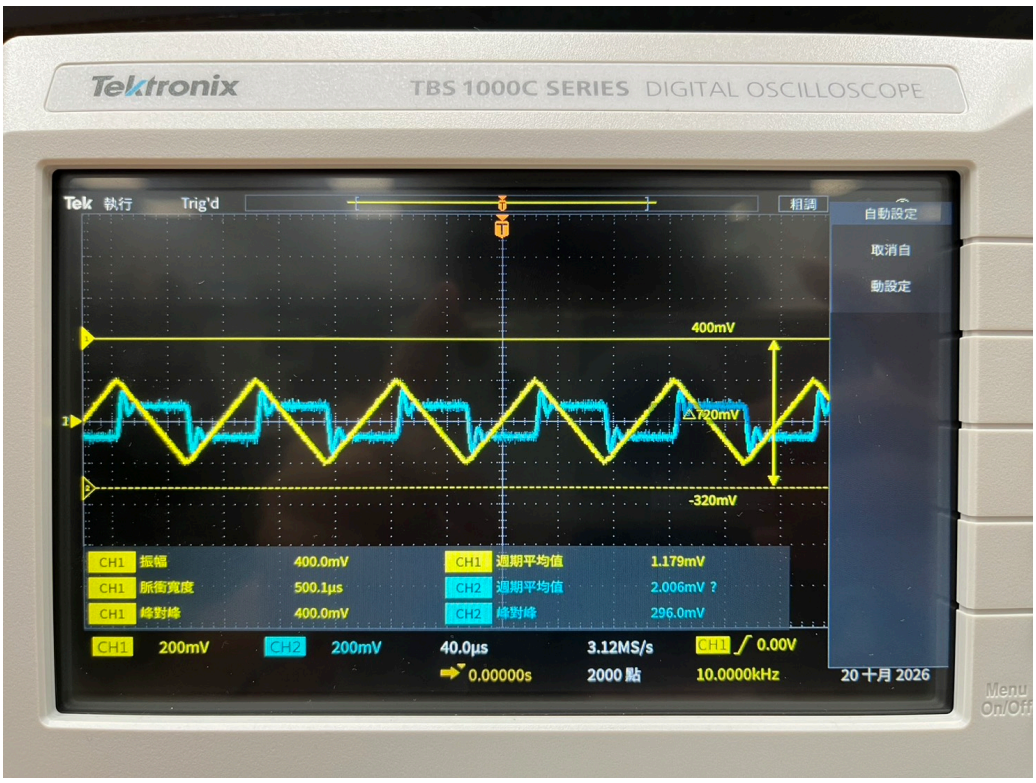
$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees lagging input with amplitude $\frac{ARC}{T}$.

$$v_O(t) = -\frac{ARC}{T}\cos\left(\frac{t}{T}\right)$$

III 實驗結果

Triangle wave:



Square wave:



Sinusoidal wave:



3 Exp. 3 Frequency Responses (I) Integrator

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator under

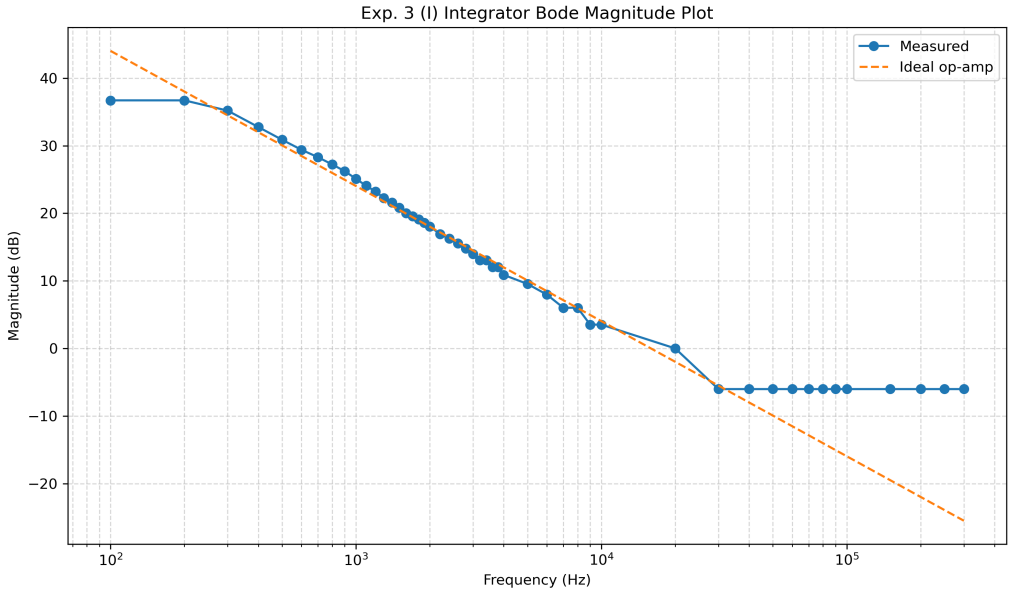
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 300\text{ kHz}$$

and sketch the Bode magnitude plot.

II　實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	27.40	36.71	63.66	44.04	-56.96
200	27.40	36.71	31.83	38.02	-13.92
300	23.00	35.19	21.22	34.49	8.38
400	17.40	32.77	15.92	32.00	9.33
500	14.00	30.88	12.73	30.06	9.96
600	11.80	29.40	10.61	28.47	11.21
700	10.40	28.30	9.09	27.13	14.35
800	9.20	27.23	7.96	25.97	15.61
900	8.20	26.24	7.07	24.95	15.92
1000	7.20	25.11	6.37	24.04	13.10
1100	6.40	24.08	5.79	23.21	10.58
1200	5.80	23.23	5.31	22.45	9.33
1300	5.20	22.28	4.90	21.76	6.19
1400	4.80	21.58	4.55	21.11	5.56
1500	4.40	20.83	4.24	20.51	3.67
1600	4.00	20.00	3.98	19.95	0.53
1700	3.80	19.55	3.74	19.43	1.47
1800	3.60	19.08	3.54	18.93	1.79
1900	3.40	18.59	3.35	18.46	1.47
2000	3.20	18.06	3.18	18.02	0.53
2200	2.80	16.90	2.89	17.19	-3.24
2400	2.60	16.26	2.65	16.43	-1.98
2600	2.40	15.56	2.45	15.74	-1.98
2800	2.20	14.81	2.27	15.09	-3.24
3000	2.00	13.98	2.12	14.49	-5.75
3200	1.80	13.06	1.99	13.93	-9.52
3400	1.80	13.06	1.87	13.41	-3.87
3600	1.60	12.04	1.77	12.91	-9.52
3800	1.60	12.04	1.68	12.44	-4.50
4000	1.40	10.88	1.59	12.00	-12.04
5000	1.20	9.54	1.27	10.06	-5.75
6000	1.00	7.96	1.06	8.47	-5.75
7000	0.80	6.02	0.91	7.13	-12.04
8000	0.80	6.02	0.80	5.97	0.53
9000	0.60	3.52	0.71	4.95	-15.18
10000	0.60	3.52	0.64	4.04	-5.75
20000	0.40	0.00	0.32	-1.98	25.66
30000	0.20	-6.02	0.21	-5.51	-5.75
40000	0.20	-6.02	0.16	-8.00	25.66
50000	0.20	-6.02	0.13	-9.94	57.08
60000	0.20	-6.02	0.11	-11.53	88.50
70000	0.20	-6.02	0.09	-12.87	119.91
80000	0.20	-6.02	0.08	-14.03	151.33
90000	0.20	-6.02	0.07	-15.05	182.74
100000	0.20	-6.02	0.06	-15.96	214.16
150000	0.20	-6.02	0.04	-19.49	371.24
200000	0.20	-6.02	0.03	-21.98	528.32
250000	0.20	-6.02	0.03	-23.92	685.40
300000	0.20	-6.02	0.02	-25.51	842.48

III Bode plot



4 Exp. 3 Frequency Responses (I) Differentiator

I Circuit

Measure frequency response of the circuit in Exp. 2 Differentiator under

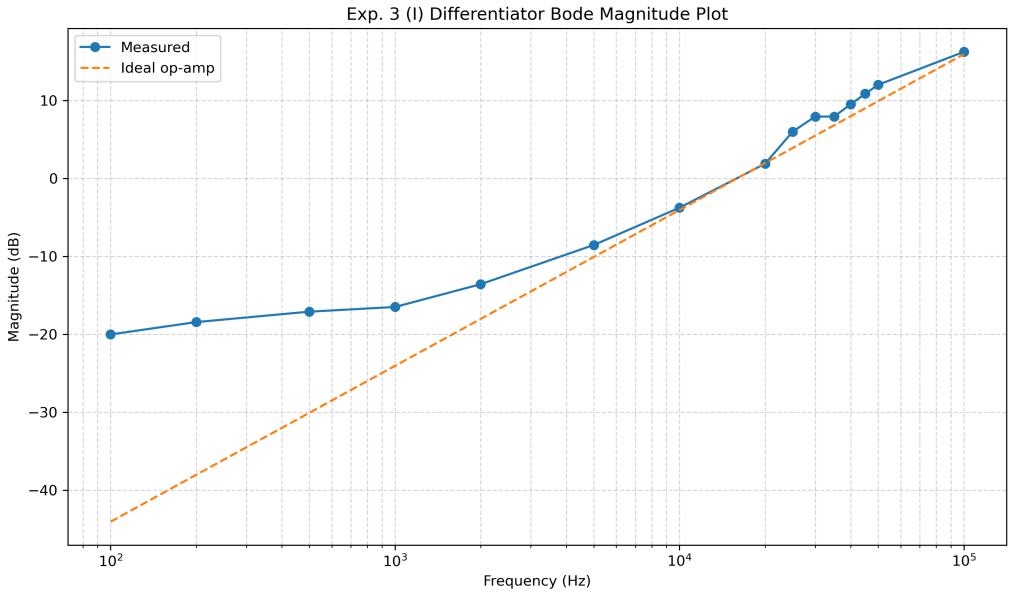
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 100\text{ kHz}$$

and sketch the Bode magnitude plot.

II 實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	0.04	-20.00	0.00	-44.04	1491.55
200	0.05	-18.42	0.01	-38.02	854.93
500	0.06	-17.08	0.01	-30.06	345.63
1000	0.06	-16.48	0.03	-24.04	138.73
2000	0.08	-13.56	0.05	-18.02	67.11
5000	0.15	-8.52	0.13	-10.06	19.37
10000	0.26	-3.74	0.25	-4.04	3.45
20000	0.50	1.90	0.50	1.98	-0.93
25000	0.80	6.02	0.63	3.92	27.32
30000	1.00	7.96	0.75	5.51	32.63
35000	1.00	7.96	0.88	6.84	13.68
40000	1.20	9.54	1.01	8.00	19.37
45000	1.40	10.88	1.13	9.03	23.79
50000	1.60	12.04	1.26	9.94	27.32
100000	2.60	16.26	2.51	15.96	3.45

III Bode plot



5 Exp. 3 Frequency Responses (II) Integrator without feedback resistor

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) under

$$v_I = 0.2 \sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

II Theoretical results

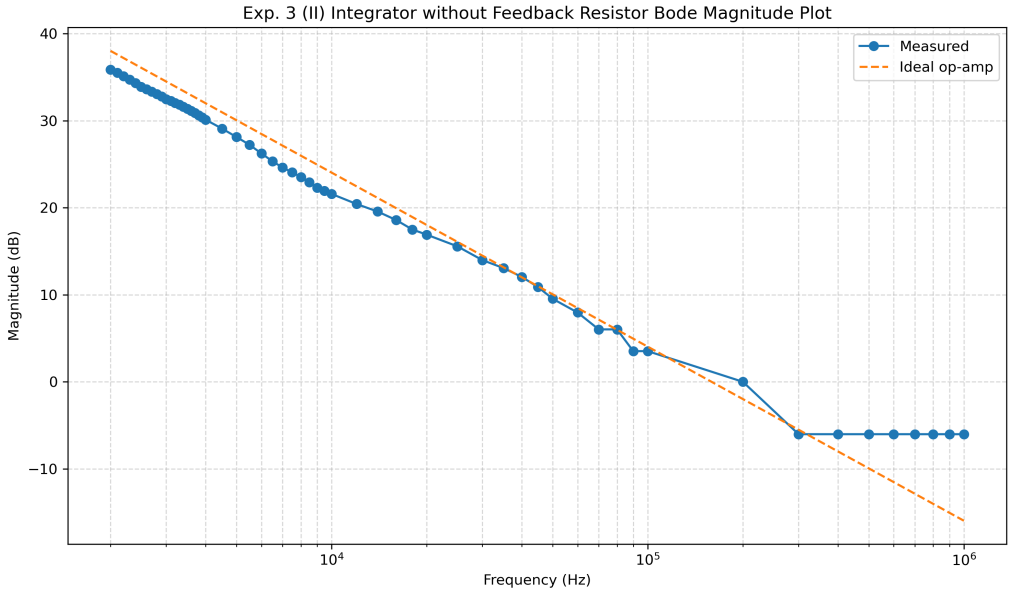
Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 159 \text{ kHz}$$

III　實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
2000	24.80	35.85	31.83	38.02	-22.09
2100	23.80	35.49	30.32	37.59	-21.49
2200	22.80	35.12	28.94	37.19	-21.21
2300	21.80	34.73	27.68	36.80	-21.24
2400	20.80	34.32	26.53	36.43	-21.59
2500	19.80	33.89	25.46	36.08	-22.25
2600	19.20	33.62	24.49	35.74	-21.59
2700	18.60	33.35	23.58	35.41	-21.11
2800	18.00	33.06	22.74	35.09	-20.83
2900	17.40	32.77	21.95	34.79	-20.74
3000	16.80	32.46	21.22	34.49	-20.83
3100	16.40	32.26	20.54	34.21	-20.14
3200	16.00	32.04	19.89	33.93	-19.58
3300	15.60	31.82	19.29	33.67	-19.14
3400	15.20	31.60	18.72	33.41	-18.82
3500	14.80	31.36	18.19	33.16	-18.63
3600	14.40	31.13	17.68	32.91	-18.57
3700	14.00	30.88	17.21	32.67	-18.63
3800	13.60	30.63	16.75	32.44	-18.82
3900	13.20	30.37	16.32	32.22	-19.14
4000	12.80	30.10	15.92	32.00	-19.58
4500	11.40	29.10	14.15	30.97	-19.42
5000	10.20	28.13	12.73	30.06	-19.89
5500	9.20	27.23	11.57	29.23	-20.52
6000	8.20	26.24	10.61	28.47	-22.72
6500	7.40	25.34	9.79	27.78	-24.44
7000	6.80	24.61	9.09	27.13	-25.23
7500	6.40	24.08	8.49	26.54	-24.60
8000	6.00	23.52	7.96	25.97	-24.60
8500	5.60	22.92	7.49	25.45	-25.23
9000	5.20	22.28	7.07	24.95	-26.49
9500	5.00	21.94	6.70	24.48	-25.39
10000	4.80	21.58	6.37	24.04	-24.60
12000	4.20	20.42	5.31	22.45	-20.83
14000	3.80	19.55	4.55	21.11	-16.43
16000	3.40	18.59	3.98	19.95	-14.55
18000	3.00	17.50	3.54	18.93	-15.18
20000	2.80	16.90	3.18	18.02	-12.04
25000	2.40	15.56	2.55	16.08	-5.75
30000	2.00	13.98	2.12	14.49	-5.75
35000	1.80	13.06	1.82	13.16	-1.04
40000	1.60	12.04	1.59	12.00	0.53
45000	1.40	10.88	1.41	10.97	-1.04
50000	1.20	9.54	1.27	10.06	-5.75
60000	1.00	7.96	1.06	8.47	-5.75
70000	0.80	6.02	0.91	7.13	-12.04
80000	0.80	6.02	0.80	5.97	0.53
90000	0.60	3.52	0.71	4.95	-15.18
100000	0.60	3.52	0.64	4.04	-5.75
200000	0.40	0.00	0.32	-1.98	25.66
300000	0.20	-6.02	0.21	-5.51	-5.75
400000	0.20	-6.02	0.16	-8.00	25.66
500000	0.20	-6.02	0.13	-9.94	57.08
600000	0.20	-6.02	0.11	-11.53	88.50
700000	0.20	-6.02	0.09	-12.87	119.91
800000	0.20	-6.02	0.08	-14.03	151.33
900000	0.20	-6.02	0.07	-15.05	182.74
1000000	0.20	-6.02	0.06	-15.96	214.16

IV Bode plot



6 Exp. 3 Frequency Responses (II) Integrator with feedback resistor

I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor C replaced with a capacitor $C = 0.001$ μF (102) and a resistor R_F in parallel connection under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

II Theoretical results

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_F = -\left(\frac{1}{R_F} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_F Cs}{R_F}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_F C}{1 + R_F Cs}v_O(0^-) - \frac{\frac{R_F}{R}}{1 + R_F Cs}V_I$$

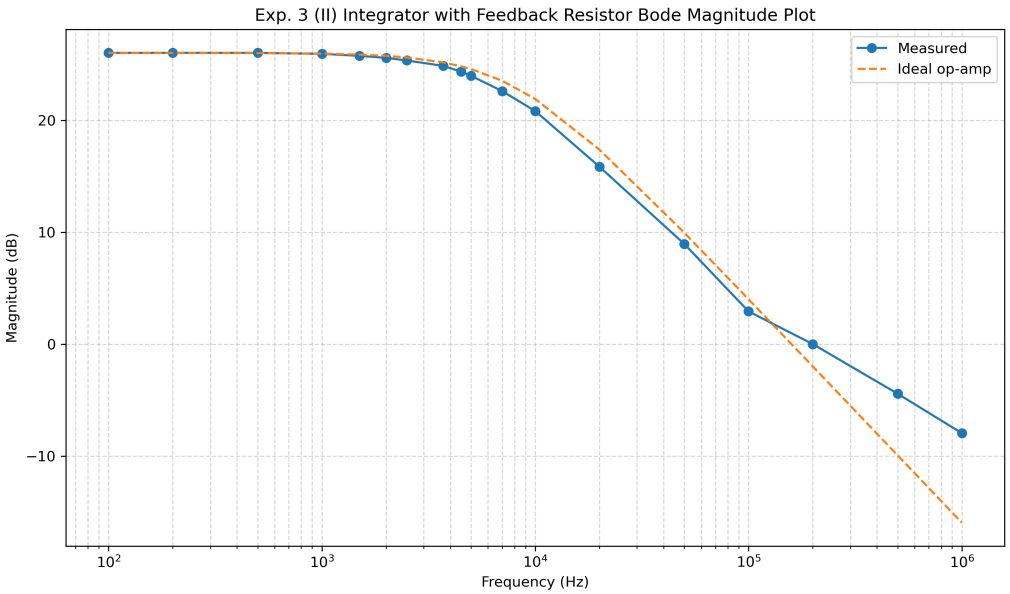
Transfer function: STC low-pass with cutoff frequency $\frac{1}{2\pi R_F C}$ and DC gain $\frac{R_F}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_F}{R}}{1 + R_F C s}$$

III 實驗結果與數據分析

frequency (Hz)	measured output voltage peak-to-peak (V)	measured gain (dB)	ideal output voltage peak-to-peak (V)	ideal gain (dB)	error w.r.t. ideal (%)
100	8.00	26.02	8.00	26.02	0.01
200	8.00	26.02	8.00	26.02	0.03
500	8.00	26.02	7.98	26.00	0.20
1000	7.92	25.93	7.94	25.95	-0.22
1500	7.76	25.76	7.86	25.87	-1.29
2000	7.60	25.58	7.76	25.75	-2.05
2500	7.40	25.34	7.63	25.61	-3.04
3700	7.00	24.86	7.25	25.17	-3.50
4500	6.60	24.35	6.96	24.82	-5.22
5000	6.32	23.97	6.77	24.58	-6.70
7000	5.40	22.61	6.01	23.53	-10.10
10000	4.40	20.83	4.98	21.91	-11.67
20000	2.48	15.85	2.96	17.38	-16.15
50000	1.12	8.94	1.26	9.95	-10.93
100000	0.56	2.92	0.63	4.01	-11.76
200000	0.40	0.00	0.32	-1.99	25.76
500000	0.24	-4.44	0.13	-9.94	88.52
1000000	0.16	-7.96	0.06	-15.96	151.34

IV Bode plot



7 誤差可能原因

The differences between the theoretical and experimental results may be attributed to the following factors:

- Non-ideal operational amplifier characteristics: The operational amplifier has finite open-loop gain, gain-bandwidth product, slew rate, etc. Thus, its behavior deviates from the ideal op-amp model, especially at high frequencies.
- Output saturation: The ideal integrator has infinite DC gain. Small DC offsets and input bias currents may make low-frequency measurements deviate from the ideal op-amp model.
- Measurement limitations: Oscilloscope precision may introduce errors, especially when the output amplitude becomes small.
- Component errors: The actual resistance and capacitance values may differ from their nominal values. Probes, wires, and contact points may have nonzero impedance.

8 實驗心得